Zohaib Ali Mughal

AI-First Product Engineer: TypeScript, Next.js, NestJS, Claude Code

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TECHNICAL SKILLS

AI-assisted development: Claude Code, Cursor, GitHub Copilot
LLM and voice AI: LLM tool calling, prompt design, Retell AI, OpenAI models, ElevenLabs TTS
Languages: TypeScript, JavaScript, Dart, SQL, Python
Frontend: Next.js, React, Tailwind CSS, TanStack Query, Radix / shadcn UI
Backend: Node.js, NestJS, Fastify, tRPC, REST / OpenAPI, Zod, BullMQ, NATS JetStream
Data and DevOps: PostgreSQL, Prisma, Supabase, Redis, Docker, AWS, Railway, Vercel, GitHub Actions
Payments and Quality: Stripe Connect, Paymob, Airwallex, Sentry, Jest, Vitest, Playwright, Testcontainers

PROFESSIONAL EXPERIENCE

Slidey LLC, Karachi **Nov 2024 – Present**
Trainee Software Engineer (Nov 2024), Software Engineer I (Mar 2025), Software Engineer II (May 2026)

ETG Platform portal.evolutiongroup.net.au | case study **Software Engineer II**

Field-service platform for an Australian electrical, security and telecom group: admin, client, supplier and technician portals. Built to stop about $5k a week lost to unbilled supplier costs and billing scheduled rather than actual hours.

- Designed a 25-package Turborepo (Next.js frontend, Fastify API, BullMQ worker): 145 tRPC routers and 1,000+ procedures over a 212-model Prisma schema (dispatch, timesheets, assets, quoting, billing); money as BigInt cents.
- Enforcement-first decision engine: one pure, unit-testable gate checks every dispatch, closeout and invoice transition, returns structured failure codes, and logs every block and manager override (8 live checks on the dispatch calendar).
- Five role-scoped portals over one API across 148 page routes: internal operations, technician mobile, client and subcontractor self-service, and tokenised public links for quotes, invoices and feedback.

Stack: Next.js, React, Fastify, tRPC, Prisma, PostgreSQL, BullMQ, Redis, Better Auth, Docker.

Plexor case study **Software Engineer I**

Multi-camera computer-vision platform for retail loss prevention: detects theft, POS under-ringing, phone use and loitering across 100+ stores and routes prioritised alerts to store and district managers.

- Led the web side: the realtime video and inference pipeline on AWS video services (about 700ms inference across 37+ detection models), alert routing on a Node.js backend, and role-scoped Next.js dashboards.

Stack: Next.js, React, Node.js, AWS.

Hyss hyss.io | case study **Software Engineer I**

Live-event staffing marketplace: companies hire certified freelance crew, with competency-matched shifts, call sheets, realtime chat, invoicing and payouts. One Next.js codebase serves web, iOS and Android through Capacitor.

- Type-safe Turborepo: 93 Prisma models, about 340 tRPC procedures, an Android Credential Manager plugin.

Stack: Next.js, React, Fastify, tRPC, Prisma, PostgreSQL, Supabase, Stripe Connect, Capacitor, Sentry.

Conek, for A-SAFE Digital asafe.com | case study **Software Engineer I**

Quote-to-cash platform for A-SAFE, integrated with their CRM, drawing and routing tools.

- Maintained and scaled event-driven microservices on NATS JetStream with type-safe contracts and clean-architecture layering; inter-service latency under 100ms.

Stack: TypeScript, Node.js, Fastify, NATS JetStream, Next.js, React, PostgreSQL, AWS.

Sportefy App Store | Play Store | case study **Trainee Software Engineer**

Sports booking and matchmaking app live in Pakistan: Flutter on both stores, Next.js admin panel, NestJS API.

Web admin panel and API

- Next.js admin dashboard over a NestJS API: 48 domain modules, 190+ versioned REST endpoints, additive-only /v1 contract with OpenAPI drift gates in CI; no release broke older installs.

Flutter app

- 21 feature modules, 823 passing tests, BLoC with clean architecture; offline request queue with replay, automatic token refresh, fault-tolerant parsing.

Stack: NestJS, Flutter, Next.js, Prisma, PostgreSQL, Supabase, Paymob, Railway, Vercel, GitHub Actions.

TECHNICAL PROJECTS

Voice AI Patient-Intake Agent github

- Phone agent that registers new patients over a U.S. number in English or Spanish: a Retell LLM agent with a 14-section prompt and 5 tool-calling functions, saving to Postgres only after the caller confirms a read-back.
- Fastify and Zod API, Drizzle on Postgres and a Next.js dashboard; the agent's tools and the REST API share one validation schema, webhooks are signature-checked, and 53 tests run against a real Postgres engine (PGlite). Deployed on Railway and Vercel.

Stack: TypeScript, Retell AI, OpenAI, Fastify, Zod, Drizzle, PostgreSQL, Next.js, Vitest, Railway, Vercel.

Training Platform Prototype **Pre-sales demo for a prospective client, 2026**

- 303-screen clickable demo of a bilingual training, credentialing and audit platform, produced in about a week: wrote the design system and screen specs, generated the screens with Claude Design, and wired 1,169 navigation rules in a React and Vite shell.
- Playwright audits clicking every wired element and checking clipping and WCAG AA contrast (4,042 clicks, 0 dead links), with parallel Claude Code agents building, reviewing and fixing in phases.

Stack: Claude Design, Claude Code, React, Vite, TypeScript, Playwright.

Automated Crypto-Derivatives Trading System

- Event-driven TypeScript monorepo (4 services, 8 shared packages): strategy processes publish signals over Redis Streams to one execution service, the only component that holds exchange keys and places orders.
- Built with an AI coding-agent workflow I set up: 7 layer-author and 4 read-only auditor agents with fenced responsibilities; 164 test files and 3,810 test cases.

Stack: TypeScript, Node.js, Fastify, tRPC, Next.js, Prisma, PostgreSQL, Redis Streams, zod, Vitest, Docker.

EDUCATION

Institute of Business Administration (IBA), Karachi **Graduated 2025**
BS Computer Science. National Talent Hunt Program Scholar (top 33 of 7,290 candidates).

ACHIEVEMENTS

- **Teaching Assistant, Mobile Application Development (IBA):** mentored 100+ students through hands-on workshops; class assessment averages rose 30%.
- **Ignite Hackathon:** 1st place for an IoT software prototype built and deployed within a 2-hour limit.
- **Trading Bazaar:** runner-up for algorithmic trading strategies.